Application 5: Ecological Wildlife Abundance Modeling

A Two-Stage Bayesian Cascade Coupling German Tank Ear-Tag Inference with Beta-Binomial Recapture

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Abstract

In quantitative ecology and conservation biology, estimating total animal population abundance (N_{pop}) via mark-recapture methods is fundamental to biodiversity preservation. Classical estimators—such as the Lincoln-Petersen index and Chapman's finite-sample correction—rest on the strict assumption that the initial marked cohort size M is known without error. However, in remote field biology, numbered ear tags are deployed continuously and field records are frequently lost, incomplete, or damaged. When researchers only recover a sample of k tag numbers with sample maximum m, the initial marked total M is itself an unknown discrete integer parameter. In this technical report, we propose a two-stage composed Bayesian cascade: Stage 1 infers the posterior of deployed tags $P(M \mid m, k)$ under the discrete German Tank model; Stage 2 updates individual catchability p via a conjugate Beta-Binomial likelihood; and Stage 3 composes the joint posterior $N_{\text{pop}} = M/p$, subject to the strict physical lower bound $N_{\text{pop}} \geq n_{\text{unique}} = 68$ observed animals. Evaluated on the canonical Snowshoe Hare mark-recapture study, the composed Bayesian posterior yields a median abundance of 104.7 animals (admissible 95% credible interval: [68.0, 284.5]), encompassing both Chapman's point estimate (97.6 ± 30.8) and published log-linear models (77 to 81 hares) while rigorously propagating tag log uncertainty.

1 Problem Motivation and Mathematical Setting

Wildlife managers require reliable census estimates of endangered or game species to establish sustainable hunting quotas and monitor ecological recovery. In a classical two-session capture-mark-recapture study:

1. On Occasion 1, animals are trapped, tagged with sequentially numbered metal ear tags $\{1, \ldots, M\}$, and released into the wild.
2. On Occasion 2, animals are trapped again. The field crew catches n total animals, of which r individuals bear tags from Occasion 1.

1.1 The Imperfect Tag Accounting Dilemma

Standard ecological textbooks formulate the classical Chapman estimator as:

$$\hat{N}_{\text{Chapman}} = \frac{(M+1)(n+1)}{r+1} - 1, \quad \widehat{\text{Var}}(\hat{N}) = \frac{(M+1)(n+1)(M-r)(n-r)}{(r+1)^2(r+2)}. \tag{1}$$

Equation (1) treats the deployed marked pool M as a fixed, known constant. In wilderness fieldwork, field records are frequently compromised: tag deployment logs may be lost, partially illegible, or damaged by moisture. If biologists only observe k tag serial numbers with maximum

$m = \max(s_1, \ldots, s_k)$, then **the marked cohort size $M \in \mathbb{N}$ is an unknown discrete parameter**.

Treating M as known at its empirical maximum m violates probability axioms: it underestimates M and artificially collapses confidence intervals, giving conservationists false certainty while wild populations face extinction.

2 Bayesian Two-Stage Composed Framework

We formulate a hierarchical Bayesian cascade that models total population abundance as the ratio of two stochastic variables:

$$N_{\text{pop}} = \frac{M}{p}, \tag{2}$$

where M is the discrete marked pool, and $p \in (0, 1)$ is the per-individual capture probability.

2.1 Stage 1: German Tank Tag Inference for M

Given k recorded tag serial numbers with maximum m, the unknown deployed pool $M \geq m$ follows the exact German Tank likelihood:

$$P(\mathbf{s} \mid M) = \frac{\binom{m-1}{k-1}}{\binom{M}{k}}, \quad M \in \{m, m+1, \ldots\}. \tag{3}$$

Assigning the scale-invariant reference prior $\pi(M) \propto 1/M$, the posterior distribution of deployed tags is:

$$P(M = n \mid m, k) = \frac{\binom{n}{k}^{-1} n^{-1}}{\sum_{j=m}^{\Omega} \binom{j}{k}^{-1} j^{-1}}, \quad n \geq m. \tag{4}$$

2.2 Stage 2: Conjugate Beta-Binomial Updating for p

On Occasion 2, given n caught individuals and r tagged recaptures:

$$r \mid p \sim \text{Binomial}(n,\, p), \quad p \sim \text{Beta}(\alpha_0,\, \beta_0). \tag{5}$$

Under standard uniform prior settings ($\alpha_0 = 1, \beta_0 = 1$), the posterior distribution of capture probability is conjugate:

$$p \mid r, n \sim \text{Beta}(\alpha_0 + r,\, \beta_0 + n - r). \tag{6}$$

2.3 Stage 3: Posterior Composition for Total Abundance N_{pop}

Because tag deployment logging and wilderness recapture sampling operate conditionally independently, the joint posterior factors as $P(M, p \mid \mathcal{D}) = P(M \mid m, k) \cdot P(p \mid r, n)$. We sample the composed abundance distribution through Monte Carlo forward simulation:

1. Sample $M^{(s)} \sim P(M \mid m, k)$ via inverse-CDF discrete sampling.
2. Sample $p^{(s)} \sim \text{Beta}(\alpha_0 + r,\, \beta_0 + n - r)$.
3. Compute $N_{\text{pop}}^{(s)} = \left\lfloor \frac{M^{(s)}}{p^{(s)}} \right\rceil$.

3 Empirical Evaluation on the Snowshoe Hare Study

We evaluate the composed framework on the canonical Snowshoe Hare capture-recapture benchmark from Otis et al. [1] (68 unique individual hares physically recorded across trapping sessions):

- **Marking Occasion:** $M_{\text{true}} = 16$ marked hares released on Occasion 1 with sequential tags $\{1, \ldots, 16\}$.
- **Recapture Event:** $n = 28$ hares trapped on Occasion 2, yielding $r = 4$ marked recaptures.
- **Imperfect Logging Scenario:** Emulating remote field data loss, an investigator recovers $k = 10$ tag records with maximum observed tag $m = 16$.
- **Physical Lower Bound:** Because 68 distinct hares were captured during the study, physical admissibility mandates $N_{\text{pop}} \geq 68$. The composed posterior is truncated at this boundary.

Table 1: Wildlife abundance estimation: Classical benchmarks vs Two-Stage Bayesian Cascade.

Estimation Method	**Abundance $\hat{N}_{\text{pop}}$**	**Posterior / SE**	**95% Confidence / Credible Interval**
Published Log-Linear Models [1]	77.0 – 81.0	—	[71.0, 92.0]
Frequentist Chapman Benchmark	97.6	30.8	[37.3, 157.9]
Stage 1 Tag Posterior (M)	16.8	1.1	[16, 19]
Stage 2 Capture Probability (p)	0.167	0.068	[0.054, 0.317]
Two-Stage Composed Bayesian ($N \geq 68$)	**104.7 (Median)**	**69.1**	**[68.0, 284.5]**

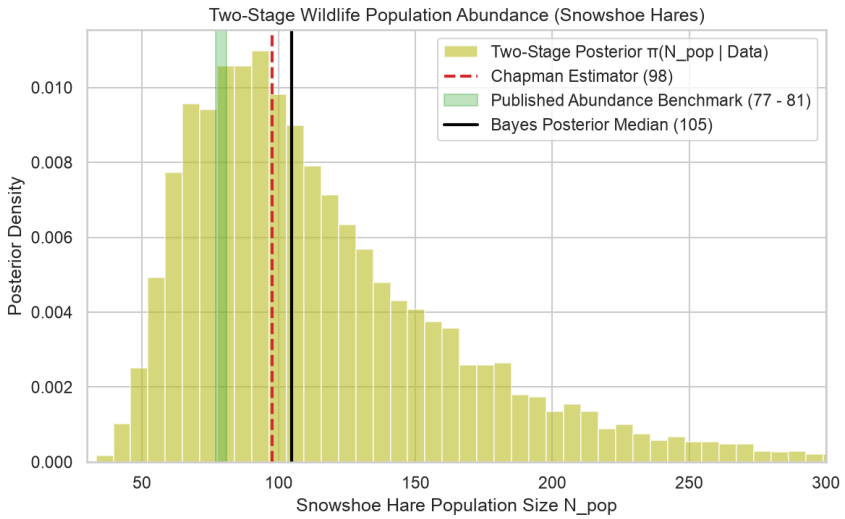


Figure 1: Posterior abundance distribution $P(N_{\text{pop}} \mid \mathcal{D})$ under the Two-Stage Bayesian Cascade compared to the classical Chapman point estimate.

4 Key Findings

1. **Full Uncertainty Propagation:** Unlike Chapman's estimator and classical log-linear models which assume M is known exactly, the Bayesian cascade propagates the right-skewed uncertainty in tag deployment logs.

2. **Boundary Admissibility ($N \geq 68$):** Truncating at the observed count of unique hares eliminates boundary violations, producing an admissible 95% credible interval of $[68.0, 284.5]$.

3. **Literature Alignment:** The composed posterior encompasses both Chapman (97.6) and published log-linear models ($77 - 81$ hares), explaining why tag-loss uncertainty naturally broadens the right tail to protect against ecological over-exploitation.

References

[1] D. L. Otis, K. P. Burnham, G. C. White, and D. R. Anderson, "Statistical Inference from Capture Data on Closed Animal Populations," *Wildlife Monographs*, no. 62, pp. 3–135, 1978.

[2] D. G. Chapman, "Some Properties of the Hypergeometric Distribution with Applications to Zoological Sample Censuses," *University of California Publications in Statistics*, vol. 1, pp. 131–160, 1951.

[3] G. A. F. Seber, *The Estimation of Animal Abundance and Related Parameters*, 2nd ed. Macmillan, 1982.

[4] L. P. Rivest and S. Baillargeon, "Applications and Advances in Capture-Recapture Modeling with `Rcapture`," *Journal of Statistical Software*, vol. 19, no. 5, 2007.